Biochemical Markers of Systemic Diseases Affecting Oral Health

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Learning Objectives

- Define biochemical markers and identify the body fluids in which they are measured.
- Recognize oral manifestations associated with common systemic diseases.
- Match important laboratory markers with the conditions they help assess.
- Explain why this information matters in dental diagnosis and treatment planning.

What Are Biochemical Markers?

Biochemical markers are measurable substances in blood, saliva, or other body fluids.

They provide information about normal body functions, inflammation, or organ damage.

They can help screen for disease, support diagnosis, and monitor treatment.

In dentistry, they help connect oral findings with a patient's general health.

Why Should Dentists Know About Them?

Some systemic diseases first produce noticeable signs in the oral cavity.

Recognizing these signs may prompt appropriate medical referral.

Laboratory results can influence dental treatment planning and precautions.

A marker supports clinical assessment; it should not be interpreted in isolation.

1. Diabetes Mellitus

Oral effects: increased risk of periodontitis, dry mouth, oral candidiasis, and delayed wound healing.

Fasting plasma glucose: measures blood glucose after fasting.

HbA1c: estimates average blood glucose over approximately 2–3 months.

Random plasma glucose: measures glucose at the time the sample is taken.

Clinical link: poor glycemic control is associated with more severe periodontal inflammation.

2. Chronic Kidney Disease

Oral effects: dry mouth, altered taste, uremic breath, mucosal changes, and increased bleeding tendency in some patients.

Serum creatinine and eGFR: help assess kidney filtration function.

Blood urea nitrogen (BUN): may rise when kidney function is reduced.

Calcium and phosphate: help assess mineral and bone disturbances in chronic kidney disease.

Dental relevance: review medical status, medications, and bleeding risk before invasive care.

3. Liver Diseases

Oral effects: jaundice of oral tissues, easy bruising, petechiae, and gingival bleeding.

ALT and AST: enzymes that may rise with liver-cell injury.

Bilirubin: increased levels can cause yellow discoloration (jaundice).

Albumin: provides information about the liver's protein-producing function.

PT/INR: helps assess blood-clotting status; important before procedures with bleeding risk.

4. Cardiovascular Diseases

Oral link: periodontal disease and cardiovascular disease share inflammatory and lifestyle risk factors.

Lipid profile: measures total cholesterol, LDL, HDL, and triglycerides to assess cardiovascular risk.

hs-CRP: a marker of low-grade systemic inflammation that can contribute to risk assessment.

Troponin: indicates heart-muscle injury, especially in suspected acute coronary syndrome.

BNP or NT-proBNP: may help assess heart failure.

5. Anemia and Nutritional Deficiencies

Oral effects: pale mucosa, smooth or sore tongue (glossitis), angular cheilitis, and recurrent ulcers.

Hemoglobin and complete blood count (CBC): help identify and classify anemia.

Ferritin: reflects iron stores; low levels commonly indicate iron deficiency.

Vitamin B12 and folate: deficiencies can cause megaloblastic anemia and oral changes.

6. Thyroid Disorders

Oral and dental effects may include altered taste, dry mouth, macroglossia, and altered tooth eruption in children.

TSH: the main initial laboratory test for many thyroid disorders.

Free T4: helps determine thyroid hormone status.

Free T3: useful in selected cases, especially suspected hyperthyroidism.

Anti-TPO antibodies: may support diagnosis of autoimmune thyroid disease.

Disease and Key Markers

Diabetes mellitus — fasting plasma glucose, HbA1c.

Chronic kidney disease — creatinine, eGFR, BUN, calcium, phosphate.

Liver disease — ALT, AST, bilirubin, albumin, PT/INR.

Cardiovascular disease — lipid profile, hs-CRP; troponin and BNP for specific clinical situations.

Anemia — CBC/hemoglobin, ferritin, vitamin B12, folate.

Thyroid disorders — TSH, free T4; selected use of free T3 and anti-TPO.

Clinical Case

A 52-year-old patient visits the dental clinic with swollen, bleeding gums and delayed healing after extraction.

The patient also reports increased thirst and frequent urination.

Question: Which systemic condition should be considered?

Clinical Case

Diabetes mellitus